EE3032 Introduction to VLSI—Midterm

1. (35 points) Answer the following questions briefly.
   (a) List the process for transferring the mask pattern to the surface of a silicon region. (3 points)
   (b) List two approaches for alleviating the crosstalk effect of two parallel interconnects. (2 points)
   (c) What silicon on insulator process is and explain how it can reduce the leakage current? (4 points)
   (d) List two methods for the oxidation in silicon. (2 points)
   (e) List two features of static CMOS circuit. (2 points)
   (f) List two components of CMOS circuit, which result in static power. (2 points)
   (g) In a logic gate, we usually implement the pull-up network with PMOS transistors and the pull-down network with NMOS transistors. Why? (2 points)
   (h) List two through-silicon via (TSV) process technologies. (2 points)
   (i) In a CMOS gate, the transistors connected to the gate output suffer from the severest body effect. Why? (2 points)
   (j) What is antenna effect? List two design methods for the fixing of antenna effect violation. (6 points)
   (k) List three bonding technologies for 3D ICs. (3 points)
   (l) The output capacitance of a CMOS logic gate in a CMOS circuit is comprised of three components. List the three components. (3 points)
   (m) List two types of FinFET structures. (2 points)
2. (15 points) (a) Design an 8:1 multiplexers using transmission gate (10 points). (b) Design a 3-input NAND gate using an 8:1 multiplexer. (5 points)
3. (15 points) Find CMOS realizations of the truth table shown in Table 1, where ABCD are the inputs and F is the output.

Table 1: Truth table.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

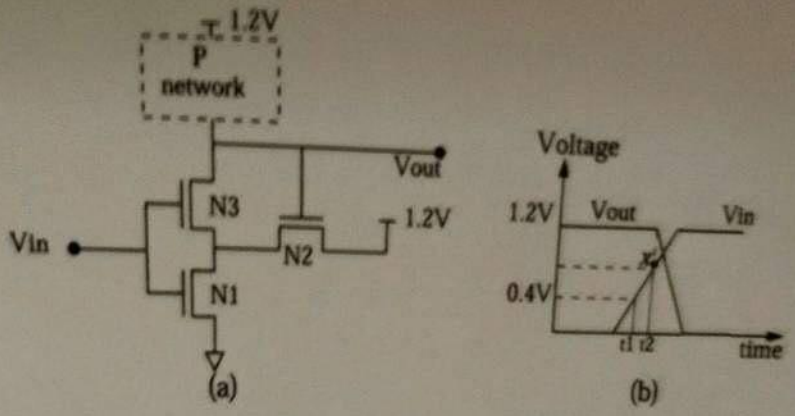


Figure 1: (a) A CMOS logic gate (b) waveforms of Vin and Vout.

4. (15 points) Fig. 1(a) shows a CMOS logic gate. Assume that the threshold voltage of the NMOS transistor is 0.4V; the turn on resistance of N1, N2, and N3 is R, 2R, and R, respectively.

   (a) If the initial voltage of the output Vout is 1.2V and an input stimulus (0→1.2V) as shown in Fig. 1(b) is applied, how much voltage of the input stimulus (Vin) at x is? (9 points)

   (b) Identify the operation region of the transistors N1, N2, and N3 from t1 to t2. (6 points)

5. (10 points) What function does the circuit shown in Fig. 2 realize? Does it have limitations?

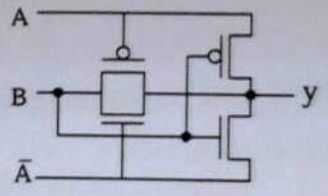


Figure 2: A CMOS circuit.

6. (10 points) Complete the following questions.

   (a) Implement the following Boolean function: $F = X(Y + Z) + XW$ using CMOS logic. Draw the schematic diagram. (5 points)

   (b) We have a CMOS logic gate realizing $F = \overline{X(Y + Z)}$ and the sizes of all the NMOS and PMOS transistors of the CMOS logic gate are W/L. Design an equivalent inverter for analyzing the rise time (t_r) and fall time (t_r) of the logic gate with input transistions XYZ: 101→100→101. (5 points)